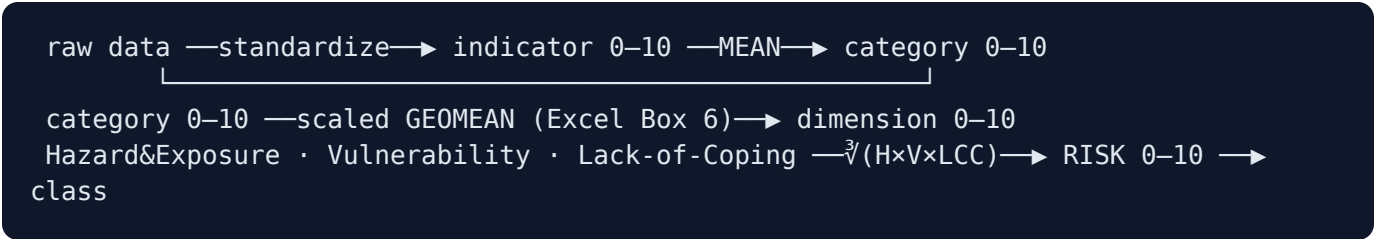


INFORM Tanzania — Methodology & Data-Integrity Manual

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Living document. Honest by design. This explains, end to end, how a raw measurement (a population count, a rainfall record, a flood event) becomes a 0–10 indicator, how indicators combine into the INFORM Risk, and how a Data-Entry edit travels the same path. Where a method is approximate, or a value looks too close to a neighbour, it is **flagged here openly** so we can improve it together — not hidden. Numbers below are real, taken from the current dataset (`src/data/tanzania-inform-risk.json`).

1. The risk journey (one picture)



Three rules, taken **exactly from the Tanzania INFORM Excel** (`TZ_INFORM_model.xlsx` , locked by `excelParity.test.js`):

Step	Method	Where
indicator → category	arithmetic MEAN	<code>AVERAGEIFS</code> in Excel
category → dimension	INFORM scaled geometric mean	Excel Box 6 (below)
dimension → risk	$\sqrt[3]{H \times V \times LCC}$ (geometric mean)	INFORM Risk index

The same three rules run in the offline build (`scripts/apply-climate-hazards.mjs`) **and** live in the app when you edit (`riskModel.applyEdits`) — so a computed value and an edited value behave identically.

2. Standardization: turning a raw number into 0–10

Every indicator must sit on the same 0–10 scale (“higher = more risk”) before it can be combined. We use **min-max** scaling across Tanzania’s districts:

$$\text{idx} = (\text{value} - \text{min}) / (\text{max} - \text{min}) \times 10$$

For quantities that are highly skewed (a few huge cities, many small rural areas) we take the **log first** (log min-max), so the scale is not dominated by one outlier:

$$\text{idx} = (\log_{10}\text{value} - \log_{10}\text{min}) / (\log_{10}\text{max} - \log_{10}\text{min}) \times 10$$

Worked example — population → Exposure (your “3900” question)

Exposure is **population density** (people per km²), log-scaled because Dar (4,600/km²) dwarfs pastoral Longido (22/km²).

- Ilala (Dar): density 4,643/km² → $\log_{10} = 3.667$ → **exposure 8.66**
- Kondoia: density 54/km² → $\log_{10} = 1.73$ → **exposure 3.19**
- Longido: density 22/km² → $\log_{10} = 1.34$ → **exposure 2.10**

So a district of (say) 3,900 people in a 1,300 km² ward = density 3.0/km² → near the bottom of the scale; the *same* 3,900 people in 5 km² = 780/km² → high. **Exposure is about concentration, not the raw headcount.** (Population comes from the NBS 2022 census, per council, summed to the district — see §7.)

Why min-max and not “rank”

An earlier draft ranked districts (1st, 2nd, ...). That spreads everything evenly but **invents differences where the data is flat** — it once put random districts top of the drought list. Min-max keeps the *real* spacing: if two districts are genuinely similar, their scores stay close (and yes, sometimes they look “too close” — that is honest, see §8).

3. Hazard indicators — what we compute, honestly

3.1 Drought (computed — CHIRPS v3 + ERA5)

Drought is **slow** and hits communities through **failed growing seasons** (→ food insecurity, IPC/MUCHALI). SPI/SPEI *frequency* is ~16 % everywhere by construction, so it can NOT rank where drought is worse. Instead we blend four real signals, each min-maxed:

$$\text{drought} = \text{minmax}(0.30 \cdot \text{aridity} + 0.20 \cdot \text{variability} + 0.15 \cdot \text{SPEI-depth} + 0.35 \cdot \text{season-failure}) \times 10$$

- **aridity** = low mean annual rainfall (1991–2024 CHIRPS v3)
- **variability** = interannual rainfall CV
- **SPEI-depth** = mean magnitude of SPEI-12 during droughts (uses ERA5 temperature → Thornthwaite PET)
- **season-failure** = how often the main 4-month rainy season falls below 80 % of normal

Result (top): Longido 9.7, Simanjiro 9.1, Singida 8.7, Hanang 8.7, Kondoa 8.6 — the real pastoral north + semi-arid centre. Bottom: Rungwe 0.5, Kyela 0.5, Mafia 1.5 — the wet highlands/islands. (*Script: `scripts/compute-drought-spi-spei.py`*.)

The computed value is **floored at the documented baseline** (`drought = max(computed, documented)`) — like flood, a known drought is never lowered, only raised where the climate evidence is stronger.

3.2 Heavy rainfall & flood (computed — CHIRPS daily + recorded events)

Two ingredients: 1. **Heavy-rain index** — count of days where the district-mean rainfall exceeds **50 mm** (heavy) and **100 mm** (very heavy), per year, 2015–2024 daily CHIRPS, min-maxed (very-heavy weighted ×2). Top: the Pemba/Zanzibar coast (Mkoani 14 heavy days/yr). 2. **Recorded floods** — a documented, *editable* event list (`data-source/flood_events.csv`) with sources (DesInventar / EM-DAT / PMO-DMD / news). `event index = min(10, 4 + 2 × #events)` .

The **physical flood hazard** = `max(heavy-rain index, event index)` — so a coastal cloudburst zone or a documented river-flood district both score high.

3.3 Flood = Hazard, amplified by Exposure (never lowered)

INFORM hazard is **Hazard × Exposure** — but exposure must only ever **raise** a flood score, never hide a flood-prone place. So:

$$\text{hazardFlood} = \max(\text{documented flood}, \text{heavy-rain index}, \text{recorded-event index})$$
$$\text{flood} = \max(\text{hazardFlood}, \sqrt{(\text{hazardFlood} \times \text{exposure})})$$

Exposure lifts the score only where population exceeds the hazard (dense flood cities); a sparsely-populated but flood-prone district keeps its full documented hazard.

- **Ilala (Dar):** heavy-rain 4.0, 2 recorded floods (2011, 2018) → event index 8 → hazardFlood = 8. Exposure 8.66 → $\sqrt{(8 \times 8.66)} = 8.3$ → flood = max(8, 8.3) = **8.3**.
- **Pangani (coastal, flood-prone but sparse):** documented hazard 9.8, exposure 2.9 → $\sqrt{(9.8 \times 2.9)} = 5.3$ → flood = max(9.8, 5.3) = **9.8 — preserved**.

⚠ An earlier draft multiplied flood *down* by exposure (Pangani 9.8 → 3.3, Korogwe 8 → 2.1 — hiding known flood areas). That bug was caught by a before/after audit and fixed: exposure now amplifies only; a documented flood hazard is never lowered.

3.4 Other hazards — deepened one by one (all raise-only)

Each is flooded at the documented baseline (`hazard_baseline.csv`) — computed/overlay evidence can raise a hazard, never lower it. - **Storms & tropical cyclone** — coastal exposure overlay graded south → north: **Cyclone Hidaya 2024** (Mafia landfall, 18,862 affected), **Lindi cyclone 1952**, Zanzibar 1872. The INFORM baseline had ~0 here — outdated after Hidaya (first cyclone to hit Tanzania). - **Heatwave (new — computed)** — ERA5 temperature climatology (hot coast/lowlands → cool highlands), capped at 6.5 (Tanzania heat is moderate, not Sahel-extreme). `heatwave_era5.csv`. - **Lightning (new — overlay)** — Lake Victoria basin, one of Earth's top flash-density hotspots (NASA LIS/OTD; ~3,000–5,000 thunderstorm deaths/yr in the basin); shore districts 8–9. - **Volcano (new — overlay)** — Ol Doinyo Lengai (active, Lake Natron), Rungwe Volcanic Province, Kilimanjaro & Meru (dormant). - **Zoonoses / plants & pests (new — overlay)** — 2020 desert-locust invasion (north), fall armyworm (maize belts), livestock disease (pastoral), tsetse (western miombo). - **Earthquake, landslide, coastal hazards, wildfire, environmental degradation** keep their **documented / INFORM SADC** values (earthquake already follows the rift; the others are differentiated). Next: wildfire from MODIS/VIIRS, earthquake refine from USGS — same overlay pattern. - **Vulnerability & coping** indicators are the next dimensions to deepen — so the Risk total will keep evolving as the model becomes more complete. That is expected and documented, not drift.

4. Vulnerability & Coping — current sources

- **Development & poverty** — NBS Household Budget Survey 2017/18 (poverty headcount); already differentiated by region (Kigoma/Geita/Kagera highest → Dar/Kilimanjaro lowest).

- **Children health & nutrition** — TDHS 2022 (under-5 stunting; Iringa/Njombe/Rukwa highest).
 - **Health conditions (NEW)** — disease burden = **HIV** (THIS 2022-23: Njombe 11.4 % → Kigoma 1.7 %) + **malaria** (TDHS-MIS 2022: Tabora 23 %, Mtwara 20 %, Lake/West high; highlands ~0), min-max blended, raise-only. `health_burden.csv` .
 - **Displaced people (NEW)** — **UNHCR 2024** refugees: Nyarugusu (Kasulu ~130 k) + Nduta (Kibondo ~75 k) → these were **0**, now **10 / 8** (burden-scaled). `displaced_refugees.csv` .
 - **Livelihoods / food security (NEW)** — **IPC / MUCHALI** Nov 2023–Oct 2024: **Longido & Monduli in Phase 3 (Crisis)**, 16 more pastoral/semi-arid councils in Phase 2 (Stressed) — raised raise-only. Only the **classified** councils get a value; the many baseline **0** s are **no-data** (INFORM SADC didn't classify them) and are left as-is, never fabricated. `food_security.csv` .
 - **WASH** — derived from **real 2022-census water points & boreholes** (more resource → lower “lack of coping”), rank-scaled 2–8.
 - **Access to health / education** — 2022 PHC facility counts.
 - **DRR implementation** — EPRP / regional EOCC / drought Anticipatory-Action records (the districts you provided) lower the institutional “lack of coping”.
 - Indicators not yet localized keep the INFORM SADC 2024 baseline.
-

5. Aggregation — the exact formulas

5.1 Indicator → Category: MEAN

`category = average of its indicators` (ignoring blanks). E.g. Kondoa Natural-hazards = `mean(drought 8.6, flood 3.8, earthquake 10, landslide 4.7, wildfire 4.75, lightning 3, zoonoses 4, heatwave 3.3, ...)` = **3.9**. *(Adding real low hazards like lightning/heat can nudge the mean — INFORM averages all hazard types; it never falls below the documented baseline.)*

5.2 Category → Dimension: INFORM scaled geometric mean (Excel Box 6)

$$\text{dimension} = (10 - \text{GEOMEAN}((10 - c_1)/10 \cdot 9 + 1, (10 - c_2)/10 \cdot 9 + 1, \dots)) / 9 \times 10$$

A low category **drags the dimension down** (geometric), unlike a plain average.

Worked example — Kondoa Hazard: Natural 3.9, Human 0.7. scaled: $(10 - 3.9)/10 \cdot 9 + 1 = 6.49$; $(10 - 0.7)/10 \cdot 9 + 1 = 9.37$. $\text{GEOMEAN} = \sqrt{(6.49 \times 9.37)} = 7.80$. $\text{dimension} = (10 - 7.80)/9 \times 10 = \mathbf{2.4}$. ✓ matches stored.

5.3 Dimension → Risk: cube root

$$\text{Risk} = \sqrt[3]{(\text{Hazard} \times \text{Vulnerability} \times \text{LackOfCoping})}$$

Kondoa: $\sqrt[3]{(2.4 \times 5.5 \times 4.6)} = \sqrt[3]{60.7} = 3.9$ (Medium). ✓

Then **Risk** is classed with the **Tanzania thresholds** (Very-Low < 2.5 < Low < 3.4 < Medium < 4.3 < High < 5.9 < Very-High).

5.4 Missing data — no-data never distorts the score

A blank indicator means **NO DATA** and is stored as `null`. Every aggregation (mean & scaled geomean) **excludes null entirely** — it is *not* read as 0: - `mean(5, 5, null) = 5` — not 3.33. **A no-data indicator can never pull a score up or down.** - `0` is different — a **real measured zero** (e.g. zero coastal-hazard exposure in landlocked Dodoma) — and **is** included. - An all-no-data category drops out cleanly (`null`), so the dimension is the scaled-geomean of the categories that actually have data.

In **Data Entry**, clearing a field sets it back to no-data (`null`) — it removes data rather than zeroing it. Each district shows a **data-coverage %** (share of the 32 leaf indicators that have data) as a reliability cue — so a score built on sparse data is flagged for what it is. The ambiguous INFORM `economic` slot is deliberately left no-data everywhere (170/170 null) and by design affects nothing. (*Locked by tests in `climateIntegrity.test.js`.*)

6. Keying data in (Data Entry) — same formulas, with provenance

1. Pick **role** (PMO/Admin apply directly; Sector officer submits for approval), region, district.
2. Edit a **dimension score** directly, or **Fill indicators** to set each 0–10 leaf — for **Hazard, Exposure, Vulnerability, Coping**.
3. **Exposure** has its own box; changing it instantly recomputes $\text{flood} = \sqrt{(\text{hazard} \times \text{exposure})}$ and the Hazard score — the live preview uses the *same* mean → scaled-geomean → $\sqrt[3]{}$ chain.
4. Choose the **Data source / authority** (NBS, TMA, MoW, MoH, MoA, PMO-DMD, DRR Coordinator, MUCHALI/IPC). Each edited indicator is **stamped** with that authority + date.
5. **Save / Submit** → **Apply**. The edit flows to the map, charts, tables. Hover the indicator and it shows “🔗 **Updated by MoW · 2026-06**”; un-edited indicators show the registry source (e.g. “🔗 **Source: TMA (+MoA, CHC) · CHIRPS v3 ...**”).

Storage today is the browser (`localStorage`); the Supabase schema is ready for multi-user.

7. Data sources (per indicator)

Indicator	Authority	Dataset	How standardized
Drought	TMA / MoA	CHIRPS v3 1991-2024 + ERA5 t2m	aridity+variability+SPEI-depth+season-fail, min-max
Flood	PMO-DMD / TMA / MoW	CHIRPS v3 daily + recorded floods	max(>50/100 mm count, events) × exposure
Exposure	NBS	2022 PHC population	density, log min-max
Poverty	NBS	HBS 2017/18	INFORM baseline
Stunting	MoH	TDHS 2022	INFORM baseline
Food security	MUCHALI / IPC (MoA)	IPC rounds	<i>to be entered</i>
WASH	MoW	2022 PHC water/boreholes	resource availability 2–8
Health/Education access	MoH / NBS	2022 PHC facilities	resource availability
DRR implementation	PMO-DMD / DRR Coordinator	EPRP/EOCC/AA records	investment overlay
Wildfire, earthquake, etc.	INFORM SADC 2024	INFORM	baseline (not yet recomputed)

8. Honest limitations — the “to-improve” list (flag & discuss)

1. **Drought exposure is agricultural, not urban.** We multiply *flood* by population exposure but **not drought** — multiplying drought by urban density would wrongly demote pastoral communities (Longido). Better: cropland/livestock exposure. *Open*.
2. **Heavy-rain now uses CHIRPS v3 daily** (2015–2024, national download complete, ~96 % day coverage). Earlier drafts used v2.0 daily; v3 gives slightly lower extreme-day counts but the same spatial pattern (Pemba/Zanzibar/Mafia coast highest).
3. **Some values sit close together** (e.g. several semi-arid districts ~8.4–8.7 drought). That is real similarity, not precision — min-max preserves true spacing rather than forcing a spread. Don't over-read 0.1 differences.
4. **Dar es Salaam drought = 10 (documented artifact).** Ilala/Temeke carry drought 10 from the documented INFORM baseline — clearly too high for a coastal city. The precautionary floor keeps documented values rather than silently lowering them, so it persists (the *computed* value is ~6.6). **Flagged for correction via Data Entry** (MoA/TMA), which now overrides it.

5. **Administrative structure (audited + reconciled). 31 regions is correct** (NBS 2022). The explorer now offers a **Council / LGA level showing the REAL 195 NBS 2022 councils** (official geodatabase boundaries, Swahili names parsed): **167 match an INFORM unit directly; the 28 new/split councils** (Kibiti ← Rufiji, Ubungo ← Kinondoni, Kigamboni ← Temeke, Mlimba/Ifakara ← Kilombero, Magharibi A/B ← Magharibi...) **inherit their parent district's data — flagged on hover** — until council-specific data exists (**never fabricated**). The INFORM **170-unit** and **31-region** levels remain. Official NBS 2022: 31 regions · 150 districts · 195 councils · 4,344 wards. Reconciliation: `scripts/export-councils.py` → `tanzania-councils.json` + `council_reconciliation.csv`.
6. **Wildfire, earthquake, lightning, storms** are still INFORM baseline — not yet physically recomputed (MODIS/VIIRS fire, USGS seismicity are the next sources).
7. **Standardization is relative to Tanzania** (min over our 150 districts), not INFORM's fixed global reference values. Fine for a national tool; note when comparing to global INFORM.
8. **Precautionary by design.** Flood and drought are *flooded* at the documented baseline — computed evidence can **raise** a hazard but never **lower** it. This errs toward flagging risk (safer for planning) at the cost of keeping a few possibly-overstated documented values (see #4). A before/after audit confirms **0 districts** had flood or risk-class lowered.
9. **Risk evolves as the model is completed — by design.** Populating previously-blank hazards (storms, heatwave, lightning, volcano, zoonoses, earthquake-refine, wildfire-refine, plus the technological hazards road-accidents & hazmat) raised High-risk districts from **18** → **43**, all raise-only (a value-audit confirms **0 districts below their documented baseline**). The coast (cyclone + heat), the Lake-Victoria zone (lightning + flood) and the western rift (earthquake) are the main risers. **Technological hazards (vehicleAccidents, hazmat) are kept deliberately moderate** — human hazards carry 50 % of the Hazard dimension, so road accidents should not dominate *disaster* risk. Deepening **vulnerability & coping** next will shift it again — toward a more complete picture, not drift. Every change is traceable to a sourced indicator.

Everything in this list is editable in Data Entry and improvable in code — that is the point of writing it down.

9. Risk over time (snapshots)

Risk is reported “**as of**” a date (`metadata.asOf` , currently the baseline). The plan to turn this into a time series — dated snapshots, **emerging** (risk ↑) vs **improving** (risk ↓) detection, and a Δ risk map — is in **RISK OVER TIME PLAN.md**. Foundation already in place: every Data-Entry edit is timestamped with its authority, the CHIRPS inputs are inherently temporal (1991–2024), and the trend-chart components exist — so future seasonal refreshes can be compared to spot communities whose risk is rising before it becomes a disaster.